

Spatial structure and productivity in US metropolitan areas

Evert J Meijers

Delft University of Technology, OTB Research Institute for the Built Environment, PO Box 5030, 2600 GA Delft, The Netherlands; e-mail: e.j.meijers@tudelft.nl

Martijn J Burger

Department of Applied Economics and ERIM, Erasmus University Rotterdam, PO Box 1738, 3000 DR Rotterdam, The Netherlands; e-mail: mburger@ese.eur.nl

Received 24 April 2009; in revised form 4 December 2009

Abstract. Recent concepts such as ‘megaregions’ and ‘polycentric urban regions’ emphasize that external economies are not confined to a single urban core, but are shared among a collection of nearby and linked cities. However, empirical analyses of agglomeration and agglomeration externalities have so far neglected the multicentric spatial organization of agglomeration and the possibility of the ‘sharing’ or ‘borrowing’ of size between cities. The authors take up this empirical challenge by analyzing how different spatial structures, in particular the monocentricity–polycentricity dimension, affect the economic performance of US metropolitan areas. Ordinary least squares and two-stage least-squares models explaining labor productivity show that spatial structure matters: polycentricity is associated with higher labor productivity. This appears to justify suggestions that, compared with more monocentric metropolitan areas, agglomeration diseconomies remain relatively limited in the more polycentric metropolitan areas, whereas agglomeration externalities are to some extent shared among the cities in such an area. However, it was also found that a network of geographically proximate smaller cities cannot substitute for the urbanization externalities of a single large city.

1 Introduction

Slowly but steadily, the image of the city has changed dramatically over the last century and a half. The industrial image of a city—as being composed of an urban core and a rural hinterland—has become increasingly obsolete. Advances in transport infrastructure and rising levels of car ownership have fuelled a process of decentralization and suburbanization, initially of people, but later followed by jobs (Boarnet, 1994). However, this trend does not stop with the establishment of new centers in the city or at its edges. We are also witnessing the ‘fusion’ of formerly relatively independent and distinct cities into wider metropolitan areas. Nowadays, what is ‘urban’ increasingly spreads out over a wider region, requiring us to think of the city as a regional phenomenon (Scott, 1988; Storper, 1997). This is reflected in the revival of the debate on city-regions (Parr, 2005) and in the conceptualization of regionalized urban entities (Florida et al, 2008; Hall and Pain, 2006; Kloosterman and Musterd, 2001). In spatial terms, the structure of such regions can be characterised as a series of towns “physically separate but functionally networked, clustered around one or more larger cities” (Hall and Pain, 2006, page 3), or “integrated sets of cities and their surrounding suburban hinterlands” (Florida et al, 2008, page 459). Therefore, the emerging spatial form of postindustrial urban regions is quintessentially polycentric (Hall, 2000; Kloosterman and Musterd, 2001; Phelps and Ozawa, 2003).

Essential to this debate on such regionalized urban entities is the underlying idea that external economies are not confined to a well-defined single urban core but, instead, are shared among a group of functionally linked settlements (Phelps and Ozawa, 2003; Sassen, 2007). The basic idea of polycentricity is that multiple centers or cities, and hence multiple sources of agglomeration economies, are colocated and

also interact, given the widening geographical scale of economic and social processes (van Oort et al, 2010). Such ‘regionalization’ of urbanization externalities has been conceptualized and described by several scholars (Coe and Townsend, 1998; Richardson, 1995), deploying terms such as ‘urban network externalities’ (Capello, 2000), ‘spatial externality fields’ (Phelps et al, 2001), or ‘regional externalities’ (Parr, 2002). Such concepts build on the concept of ‘borrowed size’, coined by Alonso (1973), who used it to explain why smaller cities that are part of a megalopolitan urban complex had much higher incomes than self-contained cities of similar size. According to Phelps and Ozawa,

“the idea of borrowed size suggests that today’s examples of megalopolitan agglomeration are based predominantly on some combination of pecuniary and technological externalities open to service industries across a group of settlements, rather than the technological externalities available at the localised scale of discrete towns or cities” (2003, page 594).

When external economies are increasingly conceptualized in relational terms (Burger et al, 2009; Gordon and McCann, 2000; Johansson and Quigley, 2004; Phelps and Ozawa, 2003), it would make sense to study agglomeration externalities at the scale of the regional urban system rather than the single city, as interactions with nearby cities may also influence the presence of agglomeration externalities. However, such research seems nonexistent. It appears that empirical analysis of economic agglomeration does not correspond to the changes in the geographical scale at which agglomeration manifests itself (Burger et al, 2010; Phelps and Ozawa, 2003). In analysis, the shape of the urban system is most often simply taken for granted (Kloosterman and Musterd, 2001; Scott, 2000). In equilibrium approaches, the economic organization of space is not a concern (see Corpataux and Crevoisier, 2007).

The principal objective of this paper is to provide some first steps to overcome the empirical deficit which characterize the current debate about the sharing of agglomeration externalities between a collection of more or less proximally located cities. Do cities actually borrow size from each other? Can a collection of nearby cities provide a substitute for the urbanization externalities of a single larger city? These key questions are at the heart of the worldwide policy debates on the geographical scale of agglomeration, be it megaregions in the USA or Asia, or the somewhat smaller—in spatial terms—polycentric urban regions that have become such a popular planning concept in Europe (see Meijers, 2005).

The approach developed in this paper to answer the question of whether cities borrow, or share, size is to include the spatial structure of metropolitan areas into empirical analysis of agglomeration externalities. Building on previous work by Anas et al (1998) and Lee and Gordon (2007), we do not limit ourselves to the monocentricity–polycentricity dimension, but also include a centralization–dispersion dimension. Labor productivity is used as a proxy for metropolitan performance. We extend existing production functions [notably those of Ciccone and Hall (1996)], with spatial variables other than the commonly used indicators such as size or average density.

The remainder of this paper is organized as follows. We synthesize the literature discussing the spatial structure of metropolitan areas in relation to their performance in section 2, which results in three testable assumptions. In section 3 we present our measurement of the spatial structure of metropolitan areas; in section 4 we give the model specification and detail other data used. In section 5 we present the estimation results, and in section 6 conclude and discuss the implications of our findings.

2 Urbanization, spatial structure, and metropolitan performance

Spatial structure has attracted increasing interest from urban geographers and planners as it is believed to affect the economic performance, environmental sustainability, and social well-being of places and their inhabitants. Clear examples are the ‘new urbanism’ and ‘smart growth’ movements in urban planning circles in the USA. Although in regional science and urban economics there is a large empirical literature which links city characteristics to urban performance, empirical work on agglomeration often does not go beyond including average density or city size (‘urbanization externalities’) as a spatially relevant factor. However, neither density nor size reveals much about a region’s spatial organization.

Indicators of the spatial organization of metropolitan areas need to address two questions (Anas et al, 1998). First, how is the urban population spread over urban centers? Second, to what extent is the metropolitan population located in urban centers or dispersed? The first question refers to a monocentric–polycentric dimension, while the second question refers to a centralization–dispersion dimension. Below, we address the relationship of size, monocentricity–polycentricity, and centralization–dispersion, respectively, with metropolitan performance, which results in three testable hypotheses.

2.1 Size and urbanization externalities

A large strand of empirical research in regional science and urban economics focuses on the benefits stemming from the size or density of the urban economy. As a general rule, these urbanization externalities can be regarded as external economies passed to firms as a result of savings from the large-scale operation of the city as a whole. These benefits are uncontrollable and unregulable for a single firm and, above all, immobile or spatially constrained (van Oort, 2004). Following Isard (1956), it is the availability of a large and multifunctional labor pool and the presence of a good infrastructure and public facilities in dense economic areas that are the sources of urbanization externalities. Relatively more urbanized areas are also more likely to accommodate universities, R&D laboratories, trade associations, and other knowledge-generating institutions. Moreover, the often diverse industry mix in an economically dense area increases the odds of interaction, generation, replication, modification, and recombination of ideas and applications across different sectors (van Oort, 2004), and protects a region from volatile demand (Frenken et al, 2007). In addition, the presence of a large internal market offers a larger degree of stability and lower transport costs (Siegel et al, 1995). However, a high degree of urbanization may also result in a dispersion of economic activities due to pollution, crime, or high land and housing prices. In this respect, one can speak of ‘urbanization diseconomies’, which are assumed to be negatively related to metropolitan performance. We expect to confirm the strong positive relation between size and metropolitan performance, but our main interest lies with the monocentricity–polycentricity and centralization–dispersion dimensions of urban spatial structure.

2.2 Monocentricity and polycentricity

Whereas metropolitan size is linked to external economies of scale, monocentricity or polycentricity, and centralization or dispersion, can be linked to external economies of structure. In the current debate on metropolitan spatial structure, the influence of monocentricity or its opposite, polycentricity, on the performance of metropolitan areas remains unclear due to a lack of empirical research, although this evidence is urgently needed (Cheshire, 2006; Kloosterman and Musterd, 2001; Lambooy, 1998; Meijers, 2008a; Parr, 2004; 2008; Turok and Bailey, 2004).

At the spatial scale of the metropolitan area, Lee and Gordon (2007) did not find that a polycentric or monocentric structure, measured by the subcenters’ share of all

center employment, affects metropolitan population or employment growth. Still, it is generally brought forward that the advantage of polycentricity at the local level is that it comes with a lack of agglomeration disadvantages (see Bertaud, 2004; Fujita et al, 1997; Goffette-Nagot and Schmitt, 1999) such as fierce competition for land and workers, congestion, and pollution exposure.

Agglomeration disadvantages appear to be confined largely to the scale of individual cities within the metropolitan area (Parr, 2002), and evidence suggests that smaller cities have a greater endogenous capacity to keep these social, economic, and environmental costs under control (Capello and Camagni, 2000). When, at the same time, the idea holds that agglomeration advantages are increasingly associated with a more regionalized spatial structure (Capello and Camagni, 2000; Parr, 2002; Sassen, 2007) as networks may substitute for agglomeration (Johansson and Quigley, 2004), then polycentricity will become an increasingly strong asset of metropolitan areas. So far, however, it remains untested whether a polycentric spatial structure in metropolitan areas leads to successful economic development (Lambooy, 1998; Parr, 2004; 2008). Building on the idea that agglomeration advantages have 'regionalized' to a considerable extent, whereas disadvantages remain local issues, we hypothesize in this paper that a more polycentric urban structure has a direct and positive effect on metropolitan performance, as it may provide a better balance between agglomeration advantages and agglomeration disadvantages.

2.2.1 *Extent of regionalization of urbanization externalities*

Despite the theoretical consensus regarding the 'regionalization' of urban externalities, the extent to which such a 'regionalization' may take place in a polycentric spatial structure is much questioned. According to Bailey and Turok (2001), the idea that the integration of separate cities results in agglomeration advantages comparable to those enjoyed by similar-sized monocentric cities is 'rather simplistic'. Parr (2004; 2008) points in this respect to the need for longer travel flows, longer commodity flows, and less convenient flows of information in polycentric urban regions. Moreover, it should be noted that "some of the advantages of urban size stem from the nature of the metropolitan environment, and are related to such factors as density, proximity, face-to-face contact, informal structures, unplanned interaction, etc" (Parr, 2004, page 236), and consequently hold less for polycentric urban regions. This is confirmed by a recent study (Meijers, 2008a) which showed that polycentric regions in the Netherlands had significantly fewer cultural, leisure, and sports amenities than did monocentric regions in which the urban population is concentrated in a single city. In this paper we test the hypothesis that polycentricity diminishes the effect of urbanization on labor productivity at the regional metropolitan scale.

2.3 **Centralization and dispersion**

A 'dispersed spatial structure' refers to the situation in which a large part of the population is not living in centers but is spread out across the territory in a non-concentrated pattern. It is not necessarily similar to urban sprawl, as this is often equated with low-density residential development, whereas dispersion concerns the issue of whether the development is taking place in centers or not, leaving aside the question of density. Whether or not dispersion negatively influences urban performance is an unsolved issue and, as far as it corresponds to urban sprawl, even controversial. According to a 2000 *Costs of Sprawl* report (Transportation Research Board, 2002), dispersed low-density development consumes more land and infrastructure, is associated with fewer fiscal impacts, whilst increasing housing costs, personal travel costs, and automobile dependence. Critics claim that spread-out, dispersed development fulfills a widespread need for safe neighborhoods, appreciating housing values, and unrestricted

use of automobiles. Glaeser and Kahn (2004) stress the association of sprawl with significant improvements in quality of life. According to the 2000 *Costs of Sprawl* study (Transportation Research Board, 2002), the alternative is a form of centralization in which development is directed to locations where it is more efficient to provide public services. This is referred to as 'smart growth'. In this study we test the hypothesis that metropolitan areas which have a higher proportion of the population living in urban places and, hence, less dispersion, perform better in terms of labor productivity.

3 Quantifying metropolitan spatial structure

A prerequisite for testing the idea of whether urbanization economies are dependent on spatial structure is to quantify this regional spatial structure. In addition to size, we distinguish two dimensions which affect spatial concentration. Here we discuss their exact measurement. The centralization–dispersion dimension refers to the extent to which population and employment is centralized in cities or dispersed over smaller nonurban places in a noncentralized pattern. The monocentricity–polycentricity dimension reflects the extent to which urban population and employment is concentrated in one city or spread over multiple cities in the wider metropolitan area (see figure 1). These two dimensions are quite similar to the two types of spatial concentration at the city level discerned by Anas et al (1998) in their classic essay on the urban spatial structure of cities (see also Lee and Gordon, 2007), although in terms of measurement we adopt them to the larger metropolitan area scale.

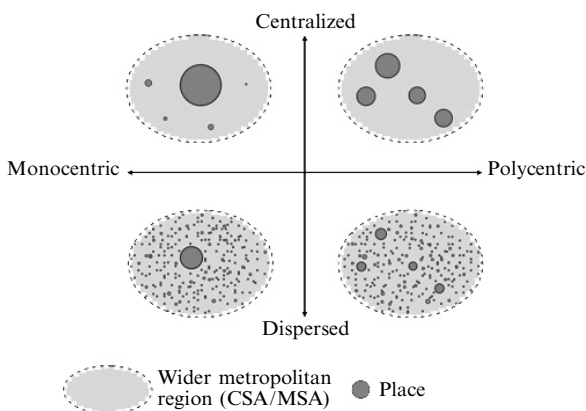


Figure 1. Dimensions of regional urban form.

3.1 Monocentricity versus polycentricity

In the literature one finds different interpretations of what makes a metropolitan area polycentric (Meijers, 2008b). Broadly speaking, there is an approach that defines polycentricity on the basis of urban morphology (see Kloosterman and Musterd, 2001; Parr, 2004), while another approach adds relational aspects, in the sense that a metropolitan area can be considered polycentric only when the cities are strongly functionally linked, which is sometimes referred to as 'relational polycentricity' or 'functional polycentricity' (see, for instance, de Goei et al, 2009; Green, 2007; Hall and Pain, 2006). Here, we adhere to the first and least restrictive vision, thus studying the monocentricity–polycentricity dimension from a morphological perspective. It is important to note that polycentricity is not so much about the presence of multiple cities in a metropolitan area, as about the balance in the size distribution of these cities. The more even this balance, and hence the less strong the hierarchy, the more polycentric the area (Kloosterman and Lambregts, 2001; Meijers, 2005; Parr, 2004).

The rank–size distribution of the regional urban system provides information on this hierarchy and is therefore a useful indication of the extent of monocentricity or polycentricity (Nordregio et al, 2004; Parr, 2004). Figure 2 presents the four largest incorporated places in the US metropolitan areas and also the regression slope that best fits the distribution of their sizes. The flatter this slope, the more polycentric the metropolitan area; conversely, the steeper this slope, the more monocentric the metropolitan area. In this example, San Antonio, TX, is obviously monocentric, whereas San Jose–San Francisco–Oakland, CA, is a clear example of a polycentric metropolitan area.

In this study we calculated the slope of the regression line of the rank–size distribution of incorporated places (cities) in each US metropolitan area.⁽¹⁾ Following a previously described method (Meijers, 2008b), we did this for different numbers of incorporated places (2, 3, and 4) per metropolitan area and then calculated the average of these three scores. The slopes found were normally distributed, meaning that the

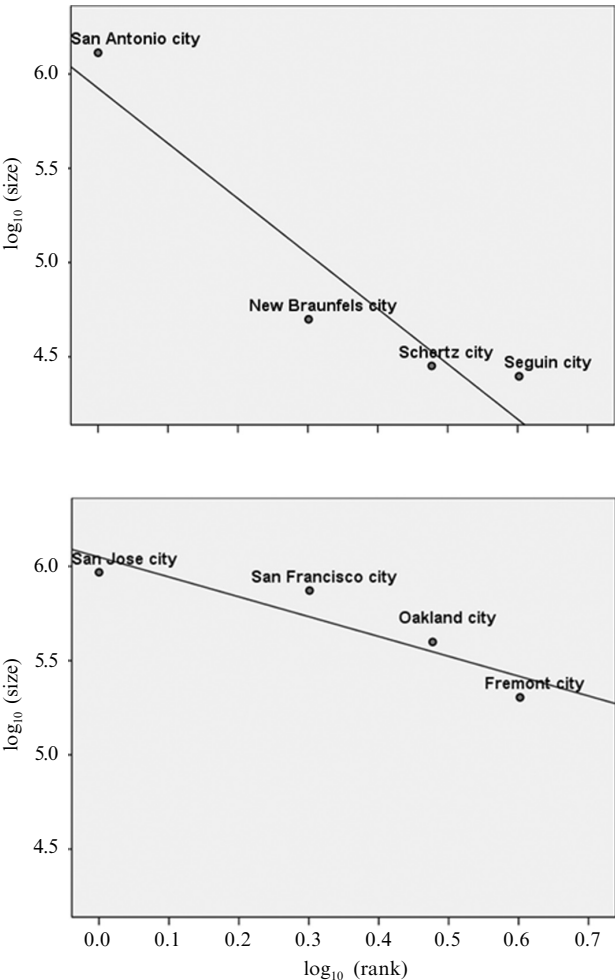


Figure 2. Rank–size distributions as measures of (a) monocentricity and (b) polycentricity.

⁽¹⁾ Considering only incorporated places of at least 5000 inhabitants in 2006. In cases where the second-largest incorporated place in the metropolitan region did not meet this threshold, we used its actual size in order to be able to calculate a slope.

majority of metropolitan areas in the USA cannot be considered very monocentric nor very polycentric, but are somewhere in between these extremes on the scale. It appears that only the most polycentric metropolitan areas can be considered polycentric urban regions in the way they are usually defined.⁽²⁾ Below, we refer to the position of a metropolitan area on this scale ranging from monocentric to polycentric as its 'degree of polycentricity', simply because higher values mean that it is more polycentric. Otherwise, we could have equally chosen to term it the 'degree of monocentricity'.

The rank–size distribution does not provide information on the spread of cities over the metropolitan territory, but polycentricity is also about spatially distinct cities (Kloosterman and Lambregts, 2001; Parr, 2004). Therefore, we need to filter out regions which score 'polycentric' in terms of the rank–size distribution, but in which the cities are part of the same contiguous built-up area. In this last case, the term 'polycentric conurbation' is more appropriate. If the two largest cities of a metropolitan area which scored as polycentric belong to the same 'urban area', which is delineated by the US Census Bureau to encompass densely settled territory, we labelled these 'polycentric conurbations'. We did not include these polycentric conurbations in the analysis as it could be argued that these metropolitan areas, although scoring as polycentric, in fact more resemble monocentric areas—the main difference being the presence of administrative boundaries dividing the built-up area (see the appendix).

3.2 Centralization versus dispersion

The share of the central city in total metropolitan area population in the USA fell rapidly from almost 64% in 1930 to 38% in 2000 (Kim, 2007). The question, then, is that of where the majority of the metropolitan population is located: are they clustered in other urban centers, or are they dispersed over the metropolitan territory in an uncentered way? In order to test the hypothesis that less centralization, and thus more dispersion, limits labor productivity, we scored each case-study area on a centralization–dispersion axis, their position being dependent on the share of the metropolitan population that was not located in urban centers of at least 25 000 inhabitants in 2006.

3.3 Association of dimensions

Both dimensions are associated in the sense that more polycentric metropolitan areas tend to be characterized slightly more often by dispersion (see table 1 for a listing of the most extreme scores on the two dimensions). There are, however, exceptions to the rule. For instance Midland–Odessa, TX, is the most polycentric metropolitan area, although it is also among the top seven most centralized areas.

4 Model, data, and estimation strategy

4.1 Model

In regional science and urban economics, many studies have investigated the effect of urbanization externalities using aggregate production functions at the city level. Although different functional forms of urban production exist, our point of departure is the models developed by Ciccone and Hall (1996) and Ciccone (2002), who use a Cobb–Douglas production function to assess the effect of agglomeration economies on localised labor productivity.

Consider the following production function of a metropolitan economy with production factors capital (K), labor (L), human capital (H), materials or intermediate

⁽²⁾ Following common definitions of such regions (see Kloosterman and Musterd, 2001; Meijers, 2005; Parr, 2004), although for a definite categorization we require more knowledge on the level of specialization and the level of interaction.

Table 1. Most monocentric, polycentric, centralized, and dispersed metropolitan regions in the USA.

Rank	Most monocentric	Most polycentric	Most centralized	Most dispersed
1	Lincoln, NE	Midland – Odessa, TX	El Paso, TX	Greenville – Spartanburg – Anderson, SC
2	Tallahassee, FL	Santa Barbara – Santa Maria, CA	Phoenix – Mesa – Scottsdale, AZ	Portland – Lewiston – South Portland, ME
3	Jacksonville, FL	Johnson City – Kingsport – Bristol (Tri-cities), TN – VA	Los Angeles – Long Beach – Riverside, CA	Columbia – Newberry, SC
4	El Paso, TX	Salt Lake City – Ogden – Clearfield, UT	Lincoln, NE	Pittsburgh – New Castle, PA
5	San Antonio, TX	Greenville – Spartanburg – Anderson, SC	San Diego – Carlsbad – San Marcos, CA	Youngstown – Warren – East Liverpool, OH – PA

inputs (M), and land (N) in which a single good is produced which is a composite of all the outputs of all producers in that metropolitan area:

$$Q = AK^{\kappa}L^{\lambda}H^{\theta}M^{\mu}N^{\nu} \; ,$$

(1)

in which Q is the nominal output. In equation (1), A represents an efficiency parameter or vector of exogenous influences on the nominal output measuring total factor productivity. Amongst others, the efficiency parameter A here reflects the urbanization externalities and metropolitan spatial structure (the degree of polycentricity and of dispersion). Given constant returns to scale ($\kappa + \lambda + \theta + \mu + \nu = 1$), equation (1) can be rewritten as

$$\left(\frac{Q}{L}\right) = A\left(\frac{K}{L}\right)^{\kappa}\left(\frac{H}{L}\right)^{\theta}\left(\frac{M}{L}\right)^{\mu}\left(\frac{N}{L}\right)^{\nu} \; ,$$

(2)

in which the output per worker (labor productivity) is a function of the capital – labor ratio, human capital – labor ratio (or, alternatively, education per worker), intermediate inputs – labor ratio, and the efficiency parameter A . By taking logarithms on both sides of equation (2), the multiplicative form can be converted into a linear stochastic form, to give the following testable equation:

$$\ln\left(\frac{Q}{L}\right) = \theta_0 + \kappa \ln\left(\frac{K}{L}\right) + \theta \ln\left(\frac{H}{L}\right) + \mu \ln\left(\frac{M}{L}\right) + \nu \ln\left(\frac{N}{L}\right) \\ + \sum_j \theta_{j+1} \ln X_j + r_j + \varepsilon \; ,$$

(3)

where equation (3) is augmented with a set of variables X with parameters θ , which are related to metropolitan size and spatial structure and enter the production function through a higher value of A (Broersma and Oosterhaven, 2009; Fogarty and Garofolo, 1988). Here, this set includes the size of the metropolitan population, the degree of polycentricity, and the degree of dispersion. Following Ciccone (2002), census region dummy variables, r_j , are included to account for remaining differences in exogenous total factor productivity, which may moderate the relationship between agglomeration,

spatial structure, and labor productivity—such as relative differences in climate, price levels, and levels of technology across metropolitan areas.

4.2 Data and variables

To estimate our production function, data were gathered for metropolitan statistical areas (MSAs), but in the many cases in which these were part of a wider combined statistical area (CSA) we used this widest possible definition of US metropolitan areas—which is based largely on commuting behavior (see the appendix). The reason to focus principally on CSAs comes from the necessity of analyzing the performance of cities in their wider spatial context, and we wanted to avoid the monocentric perspective that underlies the MSA definition. For example, it was deemed necessary for this analysis to take the San Jose–San Francisco–Oakland CSA as a unit of analysis, rather than the MSAs that make up this CSA separately. Data were gathered for those metropolitan areas located in continental USA with a total population over 250 000 inhabitants in 2006.

Our dependent variable ‘labor productivity’ was measured as the 2006 GDP in real US dollars of a metropolitan area divided by the total number of jobs in the included sectors in that area in 2006. As our research underlies a market-based model, and spatial externalities are most profound in sectors which lack exogenous endowments, agriculture, fishing, hunting, mining, and public administration were excluded (see also Brühlhart and Mathys, 2008). In addition, no figures on self-employment were available. Data on GDP by metropolitan area and sector were obtained from the US Department of Commerce’s Bureau of Economic Analysis (BEA). Data on the number of jobs were provided by the Bureau of Labor Statistics (BLS) of the US Department of Labor, and for individual sectors were taken from the 2006 American Community Survey. The new BEA estimates enable a more direct measurement of labor productivity than was previously possible, when many researchers used indirect proxies such as the mean annual wage, or household income. Descriptive statistic are provided in table 2. Note that, corresponding with specification (3), all nondummy variables in our empirical analysis are log-transformed.

Table 2. Descriptive statistics of variables ($N = 113$).

	Mean	Standard deviation	Minimum	Maximum
Labor productivity (ln)	11.29	0.197	10.70	11.85
Capital–labor ratio (ln)	11.93	0.106	11.73	12.31
Land–labor ratio (ln)	−3.661	0.732	−5.620	−1.671
Education per worker (ln)	−0.621	0.404	−1.650	0.418
Metropolitan size (ln)	13.68	0.998	12.43	16.89
Polycentricity (ln)	−0.573	0.554	−1.636	2.291
Dispersion (ln)	−0.769	0.425	−2.144	−0.124
New England	0.027	0.161	0	1
Middle Atlantic	0.071	0.258	0	1
East North Central	0.177	0.383	0	1
West North Central	0.088	0.285	0	1
South Atlantic	0.159	0.368	0	1
East South Central	0.177	0.383	0	1
West South Central	0.071	0.258	0	1
Mountain	0.088	0.285	0	1
Pacific	0.142	0.350	0	1

With respect to the factor inputs, the ‘capital–labor ratio’ within a metropolitan area was measured by linking the capital–labor ratio of broad sectors, as obtained from the Annual Survey of Manufacturers, to the industrial composition within the metropolitan area (based on a division into fifteen sectors). Hence, the obtained value is a weighted average of the capital–labor ratio across sectors. Although this operationalization does not account for regional differences within sectors (eg capital intensity, level of technology, and subsector specialization), it provides a reasonable proxy to the capital–labor ratio within metropolitan areas. The ‘land–labor ratio’ is defined as the average number of acres per worker within a metropolitan area. The human capital–labor ratio, or average ‘education per worker’, within a metropolitan area was obtained from the 2006 American Community Survey and measured as the percentage of the metropolitan population aged 25 years and older with a bachelor’s degree or higher. As our output variable is ‘gross domestic product’ (GDP), which is the total production minus intermediate inputs, we do not take intermediate goods into consideration (see also Broersma and Oosterhaven, 2009).

However, the main variables of interest in our research are the indicators related to the urbanization externalities and spatial structure. Urbanization externalities are captured by the variable ‘metropolitan size’, which is measured as the size of the metropolitan population in 2006. With respect to urban spatial structure, we include both a metropolitan area’s degree of ‘polycentricity’ and its degree of ‘dispersion’ in our model. Their measurement was discussed in section 3. Finally, dummy variables, based on the census divisions, are included; these reflect nine broad geographic regions in the United States (New England, Middle Atlantic, South Atlantic, East North Central, East South Central, West North Central, West South Central, Mountain, and Pacific; see also Lee and Gordon, 2007).⁽³⁾ These dummies attempt to control for some of the unobserved heterogeneity across metropolitan areas—in particular, differences in technology and price levels.

4.3 Endogeneity and two-stage least squares estimation (TSLS)

Although equation (3) can be estimated using conventional ordinary least squares (OLS), this estimation technique does not account for the simultaneity between metropolitan size and structure and labor productivity. In the model specification, it is assumed that metropolitan size and spatial structure have an effect on metropolitan labor productivity. However, the causality of this relationship is far from clear. On the one hand, agglomeration is often associated with a number of benefits such as labor-market pooling, accessibility to intermediate goods, knowledge spillovers, and proximity to consumers, which in turn would augment productivity. On the other hand, firms may also be attracted to economically dense areas because of the presence of higher productivity levels (Rosenthal and Strange, 2004). In other words, agglomeration can be regarded as both a cause and a consequence of labor productivity. Likewise, metropolitan spatial structure may be regarded as a consequence rather than a cause of labor productivity growth. High land and real estate prices in combination, driven by high levels of labor productivity in city centers, may disperse people and firms out of the main cities into the wider metropolitan area. In this case it is not spatial structure that directs metropolitan performance, but metropolitan performance that directs spatial structure.

Not accounting for simultaneity can lead to inconsistent estimates as this violates one of the underlying assumptions of OLS: namely, that the independent variables are uncorrelated with the disturbance term of the dependent variable.

⁽³⁾ Using census-region dummies (covering four broad geographical regions) yielded similar empirical results.

In other words, the independent variables should not be affected by the dependent variable. Probably the most common technique to deal with this (potential) simultaneity bias is to isolate the effect that runs from agglomeration and urban spatial structure to labor productivity by means of a two-stage least squares (TSLS) estimation (see also, Ciccone, 2002; Ciccone and Hall, 1996; Combes et al, 2008). In order to do so, we need instruments, or variables that are correlated with the endogenous independent variables but not with the dependent variable. Hence, they should be 'relevant', but at the same time 'exogenous'. Based on Ciccone and Hall (1996), we used five historical variables which are not related to labor productivity today, as production 60–150 years ago was organised in a very different way from today, but which did significantly contribute to today's metropolitan size and spatial structure:

- (1) the metropolitan population in 1950;
- (2) the degree of polycentricity in 1950;
- (3) the degree of dispersion in 1950;
- (4) presence or absence of a railroad in the metropolitan area in 1860 (Stover, 1961); and
- (5) agricultural land use, measured as the density of employment in agriculture outside the urban areas.

The 1950 variables are constructed in exactly the same way as our 2006 variables and their inclusion as instrumental variables is obvious. With respect to our railroad variable, it is assumed that the presence or absence of railroads in 1860 was not driven by modern productivity differences (Ciccone and Hall, 1996), but is important in explaining agglomeration at the end of the 19th century and the development of other places through suburbanization in the early 20th century. The agricultural land-use variable is used as an instrument of dispersion: more extensive agricultural land use outside the urban areas increases relative land rent as competition for land intensifies (Brueckner, 2001). This, in turn, slows down the development of housing and offices outside the urban area compared with metropolitan areas in which nonurban land is relatively cheap. Agricultural land use is, however, not related to our dependent variable as this sector was excluded in the measurement of the labor productivity of metropolitan areas.

5 Empirical results

5.1 Testing for endogeneity of metropolitan size and spatial structure

Although there may be a problem of recursive causality from a theoretical point of view, this does not necessarily mean that there is a problem from an econometric point of view (Combes et al, 2008). If indeed the independent variables of interest can be treated as exogenous, then OLS, which is a more efficient estimator, should be preferred over the TSLS estimator (Wooldridge, 2006). In order to conduct such a test, the instruments need to fulfil two general conditions: they should be relevant (not weak); and valid (exogenous).

Table 3 shows the test results on the relevance and validity of the instruments as well as the endogeneity of metropolitan size and structure. The endogenous variables were tested both separately and simultaneously. In other words, we ran three regressions in which we instrumented metropolitan size, polycentricity, or dispersion, and one regression in which we instrumented the three variables simultaneously. We included more instruments than endogenous independent variables in each specification in order to conduct an overidentification test to assess the validity of the instruments. On the basis of the Anderson canonical correlation, Cragg–Donald F -statistic, and Shea partial R^2 -statistic, we can conclude that our instruments

are relevant.⁽⁴⁾ With respect to testing the validity of the instruments, both the Sargan and Basmann tests indicate that our instruments are valid.⁽⁵⁾

Finally, we tested whether the endogenous independent variables included are also econometrically endogenous. In this, Wu – Hausman *F*-test and Durbin – Wu – Hausman χ^2 -test assess the null hypothesis that the instrumental variables are exogenous by

Table 3. First-stage results of two-stage least-squares (TSLS) regressions on metropolitan labor productivity.

	Metropolitan size	Polycentricity	Dispersion	All
<i>Instruments</i>	population in 1950; railroad in 1860	polycentricity in 1950; railroad in 1860	dispersion in 1950; agricultural land use	population in 1950; polycentricity in 1950; dispersion in 1950; railroad in 1860; agricultural land use
<i>Relevance</i>				
Anderson canonical correlation	79.28**	44.63**	35.13**	36.89**
Cragg – Donald (CD) <i>F</i> -test	114.05**	31.66**	22.11*	9.31**
Critical value CD (10% relative bias)	19.93	19.93	19.93	6.06
<i>Shea partial R²</i>				
Metropolitan size	0.70			0.75
Polycentricity		0.40		0.43
Dispersion			0.31	0.33
<i>Validity</i>				
Sargan statistic	0.29	0.08	0.00	1.00
Basmann statistic	0.25	0.07	0.00	0.86
<i>Exogeneity</i>				
Wu – Hausman <i>F</i> -test	4.78*	0.71	0.02	1.85
Durbin – Wu – Hausman χ^2 -test	5.31*	0.83	0.02	6.24
Observations	113	113	113	113
Regressors	14	14	14	14
Instruments	15	15	15	16
Excluded instruments	2	2	2	5

** $p < 0.01$, * $p < 0.05$.

⁽⁴⁾The Anderson canonical correlation statistic is significant in all four specifications, meaning that the instruments used in these specifications are adequate to identify the equation. The Cragg – Donald *F*-statistic tests whether we face a weak-instrument problem. A set of instruments is defined as ‘weak’ if the bias of the TSLS estimator, relative to the bias of the OLS estimator, exceeds the threshold of 10% (Stock and Yogo, 2005) at the 5% confidence level. As can be seen from table 3, this is the case for all specifications as the value of the Cragg – Donald *F*-statistic exceeds the critical values provided by Stock and Yogo (2005). These results are reinforced by the Shea partial *R*²-statistic, which is reasonably high for all specifications. In particular, metropolitan size appears to be well instrumented.

⁽⁵⁾In this, whether the instruments for metropolitan size and spatial structure are uncorrelated with the disturbance term is tested. As these overidentification test statistics were nonsignificant for all four specifications, we cannot reject the null hypothesis that the instruments are uncorrelated with the disturbance term of the dependent variable and can be considered valid.

comparing the TSLS estimates with the OLS estimates. From these tests it can be concluded that metropolitan spatial structure, defined by monocentricity–polycentricity and centralization–dispersion, can be treated as exogenous. In line with Lee and Gordon (2007), this is not surprising because spatial restructuring can be considered a long-term process. However, for the specification in which metropolitan size is instrumented, the Durbin–Wu–Hausman test rejects the hypothesis at a 5% confidence level ($p = 0.021$). The estimation of this equation using the OLS estimator would therefore not yield inconsistent results. Examining the specification with multiple endogenous regressors, we cannot reject the null hypothesis that metropolitan size and metropolitan spatial structure are exogenous. As both the Wu–Hausman F -test and Durbin–Wu–Hausman χ^2 -test indicate that there is no evidence for an endogenous relationship between labor productivity and metropolitan size and structure, OLS estimation should be used, given the fact that our instruments are relevant and valid and OLS the more efficient test. However, we test for robustness of our results by comparing the OLS estimates with the estimates from the TSLS specification in which only metropolitan size is treated as endogenous.

5.2 Labor productivity, urbanization externalities, and spatial structure

Model 1 in table 4 shows the results of the OLS estimation of metropolitan size and spatial structure on labor productivity, controlling for capital–labor ratio, land–labor ratio, and education, and including census division fixed effects. The White–Koenker test indicates that the null hypothesis that the residuals are homoskedastic cannot be rejected, and the Ramsey RESET test shows that we do not face an omitted variable bias problem. The VIF statistics indicate no multicollinearity problems. In general, the model fits the data well, explaining about two thirds of the variance in labor productivity across American metropolitan areas.

Table 4. Ordinary least squares (OLS) and two-stage least squares (TSLS) tests on metropolitan labor productivity, 2006, with standard errors shown in parentheses (all nondummy variables are mean-centered).

	Model 1 OLS	Model 2 TSLS ^a
Intercept	11.36 (0.110)**	11.28 (0.110)**
Capital–labor ratio (ln)	0.754 (0.232)**	0.843 (0.222)**
Land–labor ratio (ln)	0.013 (0.032)	−0.012 (0.032)
Education per worker (ln)	−0.007 (0.050)	−0.010 (0.047)
Metropolitan size (ln)	0.107 (0.020)**	0.078 (0.023)**
Polycentricity (ln) ^b	0.055 (0.023)*	0.051 (0.022)*
Dispersion (ln) ^c	0.024 (0.039)	0.030 (0.036)
Census division dummies	yes	yes
Number of observations	113	113
F -statistic	13.48	12.01
R^2 (OLS)/centered R^2 (TSLS)	0.66	0.65
Root MSE	0.12	0.12
White–Koenker (OLS)/Pagan–Hall test (TSLS)	22.7	19.8
Ramsey RESET test	0.83	0.03

** $p < 0.01$; * $p < 0.05$.
^a Instruments used in the first stage of the regression for the endogenous variable metropolitan population.
^b The higher the value, the more polycentric; the lower the value, the more monocentric.
^c The higher the value, the more dispersed; the lower the value, the more centralized.

Turning to the main results, we find a positive and significant effect of the capital–labor ratio (elasticity of 0.75) and no effect of the proportion of the workforce with a bachelor’s degree or higher on metropolitan labor productivity. However, our main interest lies in examining the effect of metropolitan size and structure on urban performance. In line with the existing empirical work on agglomeration, we find a positive and significant effect of urbanization externalities on metropolitan labor productivity. A doubling of metropolitan size increases metropolitan labor productivity by over 10%. This is slightly higher than the consensus view—that a doubling of city size increases productivity by between 3% and 8% (Rosenthal and Strange, 2004)—and the average of 5.4% found by Melo et al (2009) in a meta-analysis of such estimates.

With respect to metropolitan spatial structure, we find mixed results in the sense that we find a positive and significant effect of the degree of polycentricity on metropolitan labor productivity, but no effect of the degree of dispersion. We had expected that dispersion would have a negative effect on labor productivity, but this is not the case. However, the positive effect of the degree of polycentricity on metropolitan labor productivity confirms our theoretical expectation. A doubling of the degree of polycentricity increases metropolitan labor productivity by 5.5%. This means that the labor productivity in metropolitan areas in which the urban population is relatively evenly spread over multiple places in the metropolitan area leads to higher labor productivity than is found in metropolitan areas in which the urban population is concentrated in one large city, holding everything else constant. We elaborate on this important finding in the concluding section.

Comparing the OLS and TSLS estimator (model 2 in table 4), it can be seen that the urbanization externalities effect is over 25% less when estimated using TSLS (elasticity of 0.078), while the effect of the capital–labor ratio is larger. The parameter estimates of polycentricity and dispersion do not significantly differ across the models.

Examining the two interaction effects between metropolitan structure and urbanization externalities (models 3 and 4 in table 5), the only negative effect we found is

Table 5. Ordinary least squares (OLS) and two-stage least squares (TSLS) tests on metropolitan labor productivity, 2006—interaction effects, with standard errors shown in parentheses (all nondummy variables are mean-centered).

	Model 3 OLS	Model 5 TSLS ^a
Intercept	11.33 (0.110)** a	11.26 (0.108)**
Capital–labor ratio (ln)	0.752 (0.229)**	0.824 (0.216)**
Land–labor ratio (ln)	0.002 (0.031)	−0.018 (0.031)
Education per worker (ln)	0.015 (0.049)	0.015 (0.045)
Metropolitan size (ln)	0.094 (0.021)**	0.072 (0.023)**
Polycentricity (ln)	0.037 (0.023)	0.033 (0.022)#
Dispersion (ln)	0.038 (0.039)	0.047 (0.037)
Metropolitan size × polycentricity	−0.059 (0.021)**	−0.063 (0.020)**
Metropolitan size × dispersion	−0.008 (0.024)	−0.016 (0.023)
Census division dummies	yes	yes
Number of observations	113	113
F-statistic	13.48	12.11
R ² (OLS)/centered R ² (TSLS)	0.68	0.68
Root MSE	0.12	0.11
White–Koenker (OLS)/Pagan–Hall test (TSLS)	21.11	20.32
Ramsey RESET test	0.07	1.09

** $p < 0.01$; * $p < 0.05$; $p < 0.10$.

^a Instruments used in the first stage of the regression for endogenous variable metropolitan population.

for the interaction between metropolitan size and polycentricity on metropolitan labor productivity. The negative and significant interaction term can be interpreted as due to the fact that urbanization externalities are larger in monocentric cities. This confirms our theoretical assumption that polycentricity diminishes the effect of size on labor productivity. We discuss the important theoretical implications of this finding further in the next section. Similarly, it can also be interpreted as the benefits of polycentricity holding more for smaller regions than for large regions (in terms of population).

6 Conclusion and discussion

6.1 Main findings

Recent theoretical interest among geographers has turned to postindustrial forms of urban agglomeration. The industrial image of a metropolitan area as being composed of an urban core and a rural hinterland is in many cases becoming obsolete and appears to be being replaced by increasingly polycentric metropolitan areas which spread over larger territories, thereby including multiple cities which together constitute a metropolitan network of cities. The 'city' is becoming a regional phenomenon, and calls have been made that this should also manifest itself in the study of agglomeration economics as these seem to be associated also with wider, and more polycentric, forms of metropolitan agglomeration rather than being confined to a single urban core (Alonso, 1973; Cheshire, 2006; Kloosterman and Musterd, 2001; Parr, 2008; Phelps and Ozawa, 2003). Despite these calls, however, several scholars have suggested that empirical research on agglomeration externalities in relation to the regional urban system is rather nonexistent, unlike to the more local scale of polycentric cities (Lee and Gordon, 2007). An important reason for this empirical deficit is that many existing empirical analyses of agglomeration still originate from definitions of metropolitan areas which depart from a single nodal perspective. By this, we mean that the vast majority of countries define functional urban areas as being one central city and its hinterland, thereby ignoring the possibility of having multiple core cities. Moreover, particularly in urban economics, spatial structure is hardly an issue and is often approximated by general indicators such as size or density, which do not reveal much detail of the spatial structure within regions.

In this paper we take up this empirical challenge by analyzing how different spatial structures affect the development of agglomeration externalities in US metropolitan areas in 2006 with over 250 000 inhabitants. Externalities were measured in terms of labor productivity. The widest official delimitation of metropolitan areas (CSAs) was used, and these were supplemented with MSAs which do not form part of such CSAs.

As regards spatial structure, we assumed that, as well as size, two dimensions are important as these affect urban concentration within the region: (1) the monocentricity–polycentricity dimension which indicates the extent to which the urban population is concentrated in one urban centre (monocentricity), or spread over multiple urban centres in a metropolitan area in a balanced way (polycentricity); and, (2) the centralization–dispersion dimension, which indicates the extent to which the metropolitan population is concentrated in centers. In addition to these indicators relating to spatial structure, we controlled for the capital–labor ratio, land–labor ratio, and human capital.

The empirical analysis in this paper resulted in four findings concerning the relationship between a region's spatial structure and its economic performance in terms of labor productivity.

(1) *Metropolitan areas with more dispersion do not perform worse in terms of labor productivity.*

Whether a large part of the population is concentrated in urban centers or lives outside of these centers in the nonurban part of the metropolitan area was not found to influence labor productivity. Thus we found no evidence for the expectation that dispersion is harmful to labor productivity.

(2) *Metropolitan areas which are more polycentric show higher labor productivity.*

We found a positive and significant direct effect of the degree of polycentricity on labor productivity. The more evenly the population is spread across the different places in a metropolitan area, the higher the labor productivity in that metropolitan area. As labor productivity proxies the balance between agglomeration economies and diseconomies, we can conclude that this balance is better in the more polycentric metropolitan areas. This can be explained by the assumption that urbanization diseconomies are less in the more polycentric areas, and the idea that urbanization economies have 'regionalized' to some extent whereas 'urbanization diseconomies' appear to be confined to the city boundaries. So this result appears to confirm ideas that agglomeration externalities spread over larger distances, and may interact in regions where multiple urban places, and hence multiple source of agglomeration externalities, are colocated. As such, it confirms that agglomeration economies need to be conceptualized in relational terms. Thus, as Phelps and Ozawa (2003) proposed, external economies are not confined to a single urban core but, instead, appear to be shared among a group of functionally linked settlements. This last appears to be the case for the relatively more polycentric metropolitan areas. Finding 3, however, qualifies the extent to which this 'regionalization' of agglomeration economies takes place.

(3) *The effect of metropolitan size decreases as metropolitan areas become more polycentric.*

Metropolitan size tends to reflect the presence of urbanization externalities. It was found that the degree of polycentricity has an indirect effect on labor productivity as urbanization externalities are fewer in the more polycentric metropolitan areas. Put differently, the productivity gains of size diminish as the metropolitan area becomes more polycentric. These results are in line with a previous finding (Meijers, 2008a), that the more polycentric a region was, the fewer cultural, leisure, and sports amenities were present. It also confirms the doubts raised by Parr (2004; 2008) and Bailey and Turok (2001) that the magnitude of urbanization externalities in a polycentric metropolitan area is less compared with a monocentric metropolitan area. From a theoretical perspective, we cannot expect the advantages based on density, proximity, and the easy and sometimes unplanned exchange of information to be equally present in a metropolitan area in which population is spread over multiple smaller cities as in one where the population is concentrated in a single large city. This result means that a collection of cities does not provide a substitute for the urbanization externalities of a single large city, even though the size of the population in both metropolitan areas may be similar.

(4) *Polycentricity appears to be more beneficial in smaller metropolitan areas.*

Our findings on the interaction effect could also be interpreted as indicating that the positive effect of having a relatively more polycentric spatial structure on labor productivity found here decreases as the population size of the metropolitan area increases. A possible reason for polycentricity being relatively more beneficial in smaller metropolitan areas is that cities in smaller polycentric metropolitan areas may be more

functionally related than those in larger polycentric metropolitan areas. This needs to be tested, however.

6.2 Research agenda

More generally, the relation between the strength and extent of functional linkages between cities in a metropolitan area and their performance as a regional urban system deserves further exploration. Also, external linkages between more distant metropolitan regions require our attention. The idea that such external linkages are of great importance in explaining metropolitan performance appears to hold in particular for the largest metropolitan areas, as these are the main locations for knowledge-intensive activities such as advanced producer services and, as such, the centers in the global network of information and knowledge exchange (Hall and Pain, 2006; Sassen, 1991). In this respect, it would be interesting to see whether agglomeration externalities are also shared between even more distant cities. Other important research issues following from our findings concern whether the gap in urbanization externalities between more polycentric and more monocentric metropolitan areas is increasing or diminishing with time, whether typical agglomeration disadvantages are indeed less severe in polycentric metropolitan areas, and whether the effects of spatial structure on productivity levels differ across sectors.

Acknowledgements. Evert Meijers wishes to thank the funding provided by the Netherlands Organisation for Scientific Research (NWO). The paper has benefited from well thought-out comments by Nick Phelps as well as three anonymous reviewers.

References

- Alonso W, 1973, "Urban zero population growth" *Daedalus* **109** 191–206
- Anas A, Arnott R, Small K A, 1998, "Urban spatial structure" *Journal of Economic Literature* **36** 1426–1464
- Bailey N, Turok I, 2001, "Central Scotland as a polycentric urban region: useful planning concept or chimera?" *Urban Studies* **38** 697–715
- Bertaud A, 2004, "The spatial organization of cities: deliberate outcome or unforeseen consequence?", WP 2004-01, Institute of Urban and Regional Development, Berkeley, CA
- Boarnet M G, 1994, "An empirical model of intermetropolitan population and employment growth" *Papers in Regional Science* **73** 135–152
- Broersma L, Oosterhaven J, 2009, "Regional labour productivity in The Netherlands: evidence of agglomeration and congestion effects" *Journal of Regional Science* **49** 483–511
- Brueckner J K, 2001, "Urban sprawl: lessons from urban economics", in *Brookings – Wharton Papers on Urban Affairs* Eds W G Gale, J Rothenberg Pack (Brookings Institution Press, Washington, DC) pp 65–89
- Brühlhart M, Mathys N A, 2008, "Sectoral agglomeration economies in a panel of European regions" *Regional Science and Urban Economics* **38** 348–362
- Burger M J, van Oort F G, Frenken K, van der Knaap B, 2009, "Networks and economic agglomerations" *Tijdschrift voor Economische en Sociale Geografie* **100** 139–144
- Burger M J, van Oort F G, van der Knaap B, 2010, "A treatise of the geographical scale of agglomeration externalities and the modifiable areal unit problem" *Scienze Regionali* forthcoming
- Capello R, 2000, "The city network paradigm: measuring urban network externalities" *Urban Studies* **37** 1925–1945
- Capello R, Camagni R, 2000, "Beyond optimal city size: an evaluation of alternative urban growth patterns" *Urban Studies* **37** 1479–1496
- Cheshire P C, 2006, "Resurgent cities, urban myths and policy hubris: what we need to know" *Urban Studies* **43** 1231–1246
- Ciccone A, 2002, "Agglomeration effects in Europe" *European Economic Review* **46** 213–227
- Ciccone A, Hall R E, 1996, "Productivity and the density of economic activity" *American Economic Review* **86** 54–70
- Coe N M, Townsend A R, 1998, "Debunking the myth of localized agglomerations: the development of a regionalized service economy in south-east England" *Transactions of the Institute of British Geographers, New Series* **23** 385–404

- Combes P P, Mayer T, Thisse J F, 2008 *Economic Geography: The Integration of Regions and Nations* (Princeton University Press, Princeton, NJ)
- Corpataux J, Crevoisier O, 2007, "Economic theories and spatial transformations clarifying the space – time premises and outcomes of economic theories" *Journal of Economic Geography* **7** 285 – 309
- de Goei B, Burger M J, Kitson M, van Oort F G, 2009, "Functional polycentrism and urban network development in the greater South East UK: evidence from commuting patterns" *Regional Studies* forthcoming
- Florida R, Gulden T, Mellander C, 2008, "The rise of the mega-region" *Cambridge Journal of Regions, Economy and Society* **1** 459 – 476
- Fogarty M, Garofolo G, 1988, "Urban spatial structure and productivity growth in the manufacturing sector of cities" *Journal of Urban Economics* **23** 60 – 70
- Frenken K, van Oort F G, Verburg T, 2007, "Related variety, unrelated variety and economic growth" *Regional Studies* **41** 685 – 697
- Fujita M, Thisse J F, Zenou Y, 1997, "On the endogenous formation of secondary employment centres in a city" *Journal of Urban Economics* **41** 337 – 357
- Glaeser E L, Kahn M E, 2004, "Sprawl and urban growth", in *Handbook of Regional Science and Urban Economics, Volume 4: Cities and Geography* Eds J V Henderson, J-F Thisse (Elsevier, Amsterdam) pp 2481 – 2527
- Goffette-Nagot F, Schmitt B, 1999, "Agglomeration economies and spatial configurations in rural areas" *Environment and Planning A* **31** 1239 – 1257
- Gordon I, McCann P, 2000, "Industrial clusters: complexes, agglomeration and/or social networks?" *Urban Studies* **37** 513 – 532
- Green N, 2007, "Functional polycentricity: a formal definition in terms of social network analysis" *Urban Studies* **44** 2077 – 2103
- Hall P, 2000, "Global city-regions in the twenty-first century", in *Global City Regions: Trends, Theory, Policy* Ed. A J Scott (Oxford University Press, Oxford) pp 59 – 77
- Hall P, Pain K, 2006 *The Polycentric Metropolis: Learning from Mega-city Regions in Europe* (Earthscan, London)
- Hohenberg P, Lees L, 1985 *The Making of Urban Europe, 1000 – 1994* (Harvard University Press, Cambridge, MA)
- Isard W, 1956 *Location and Space Economy* (MIT Press, Cambridge, MA)
- Johansson B, Quigley J M, 2004, "Agglomeration and networks in spatial economies" *Papers in Regional Science* **83** 165 – 176
- Kim S, 2007, "Changes in the nature of urban spatial structure in the United States, 1890 – 2000" *Journal of Regional Science* **47** 273 – 287
- Kloosterman R C, Lambregts B, 2001, "Clustering of economic activities in polycentric urban regions: the case of the Randstad" *Urban Studies* **38** 717 – 732
- Kloosterman R C, Musterd S, 2001, "The polycentric urban region: towards a research agenda" *Urban Studies* **38** 623 – 633
- Lambooy J G, 1998, "Polynucleation and economic development: the Randstad" *European Planning Studies* **6** 457 – 466
- Lee B, Gordon P, 2007, "Urban spatial structure and economic growth in US metropolitan areas", paper presented at the 46th annual meeting of the Western Regional Science Association, Newport Beach, CA, http://www-rcf.usc.edu/~pgordon/pdf/structure_growth_200701.pdf
- Meijers E J, 2005, "Polycentric urban regions and the quest for synergy: is a network of cities more than the sum of the parts?" *Urban Studies* **42** 765 – 781
- Meijers E J, 2008a, "Summing small cities does not make a large city: polycentric urban regions and the provision of cultural, leisure and sports amenities" *Urban Studies* **45** 2323 – 2345
- Meijers E J, 2008b, "Measuring polycentricity and its promises" *European Planning Studies* **16** 1313 – 1323
- Melo P C, Graham D J, Noland R B, 2009, "A meta-analysis of estimates of urban agglomeration economies" *Regional Science and Urban Economics* **39** 332 – 342
- Nordregio (lead partner) (Ed.), 2004, "ESPON 1.1.1: potentials for polycentric development in Europe", project report, Nordregio/ESPON Monitoring Committee, Stockholm/Luxembourg
- OMB, 2000 *Standards for Defining Metropolitan and Micropolitan Regions* Federal Register 65, No. 249, Office of Management and Budget, Washington, DC
- Parr J B, 2002, "Agglomeration economies: ambiguities and confusions" *Environment and Planning A* **34** 717 – 731
- Parr J B, 2004, "The polycentric urban region: a closer inspection" *Regional Studies* **38** 231 – 240

-
- Parr J B, 2005, "Perspectives on the city-region" *Regional Studies* **39** 555–566
- Parr J B, 2008, "Cities and regions: problems and potentials" *Environment and Planning A* **40** 3009–3026
- Phelps N A, Ozawa T, 2003, "Contrasts in agglomeration: proto-industrial, industrial and post-industrial forms compared" *Progress in Human Geography* **27** 583–604
- Phelps N A, Fallon R J, Williams C L, 2001, "Small firms, borrowed size and the urban–rural shift" *Regional Studies* **35** 613–624
- Richardson H W, 1995, "Economies and diseconomies of agglomeration", in *Urban Agglomeration and Economic Growth* Ed. H Giersch (Springer, Berlin) pp 123–155
- Rosenthal S S, Strange W C, 2004, "Evidence on the nature and sources of agglomeration economies", in *Handbook of Regional Science and Urban Economics, Volume 4: Cities and Geography* Eds J V Henderson, J-F Thisse (Elsevier, Amsterdam) pp 2119–2171
- Sassen S, 1991 *The Global City: New York, London, Tokyo* (Princeton University Press, Princeton, NJ)
- Sassen S, 2007, "Megaregions: benefits beyond sharing trains and parking lots?", in *The Economic Geography of Mega-regions* Ed. K S Goldfield (The Policy Research Institute for the Region, Princeton, NJ) pp 59–83
- Scott A J, 1988 *Metropolis: From the Division of Labor to Urban Form* (University of California Press, Berkeley, CA)
- Scott A J, 2000, "Economic geography: the great half-century" *Cambridge Journal of Economics* **24** 483–504
- Siegel P, Johnson T, Alwang J, 1995, "Regional economic diversity and diversification" *Growth and Change* **26** 261–284
- Stock J H, Yogo M, 2005, "Testing for weak instruments in linear IV regression", in *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg* Eds D W K Andrews, J H Stock (Cambridge University Press, Cambridge) pp 80–108
- Storper M, 1997 *The Regional World: Territorial Development in a Global Economy* (Guilford Press, New York)
- Stover J F, 1961 *American Railroads* (The University of Chicago Press, Chicago, IL)
- Transportation Research Board, 2002 *Costs of Sprawl—2000* (National Academy Press, Washington, DC)
- Turok I, Bailey N, 2004, "The theory of polynuclear urban regions and its application to central Scotland" *European Planning Studies* **12** 371–389
- van Oort F G, 2004 *Urban Growth and Innovation, Spatially Bounded Externalities in the Netherlands* (Ashgate, Aldershot, Hants)
- van Oort F G, Burger M J, Raspe O, 2010, "On the economic foundation of the urban network paradigm: spatial integration, functional integration and economic complementarities within the Dutch Randstad" *Urban Studies* **47** 725–748
- Woolridge J M, 2006 *Introductory Econometrics: A Modern Approach* (Thomson South-Western Educational Publishing, Mason, OH)

Appendix

Selection of metropolitan regions

In principle, all CSAs as well as MSAs not part of a CSA with a total population of more than 250 000 inhabitants in 2006 located in the US continental states were included in the analysis. Metropolitan areas are defined by the US Office of Management and Budget (OMB), and the 2007 revised version of the 2000 definitional standards was used. An MSA contains a core urban area with a population of 50 000 or more. It consists of one or more counties, and includes the counties containing the core urban area as well as any adjacent counties of which more than 25% of employed residents work in the urban core. MSAs which are adjacent may be joined in order to form a Combined Statistical Area if the employment interchange is at least 25. Adjacent MSAs which have an employment interchange measure of at least 15 and less than 25 are combined if local opinion favors combination (OMB, 2000). There are also 'micropolitan' areas which contain an urban core with a population of at least 10 000 but less than 50 000. These are not considered here, unless they form part of a CSA that is also composed of at least one MSA.

Some regions had to be left out for other reasons:

- (1) CSAs are composed of metropolitan statistical areas and/or micropolitan statistical areas. Often, data were not available for micropolitan areas. In a limited number of cases, this meant that no reliable data for the CSA could be retrieved. There were two CSAs in which more than 20% of their population was located in a micropolitan statistical area; these were left out for this reason. These were Charlotte–Gastonia–Salisbury, NC–SC; Fort Wayne–Huntington–Auburn, IN; and Lexington–Fayette–Frankfort–Richmond, KY.
- (2) New Orleans was left out as the data appeared biased as a result of Hurricane Katrina.

As argued in the text, polycentric conurbations had to be left out. These include:

Albany–Schenectady–Amsterdam, NY;
 Charleston–North Charleston, SC;
 Dallas–Fort Worth, TX;
 Davenport–Moline–Rock Island, IA–IL;
 Fayetteville–Springdale–Rogers, AR–MO;
 Hartford–West Hartford–East Hartford, CT;
 McAllen–Edinburg–Mission, TX;
 Miami–Fort Lauderdale–Miami Beach, FL;
 Minneapolis–St. Paul–St. Cloud, MN–WI;
 Palm Bay–Melbourne–Titusville, FL;
 Provo–Orem, UT;
 Sarasota–Bradenton–Punta Gorda, FL;
 Scranton–Wilkes–Barre, PA;
 Tampa–St. Petersburg–Clearwater, FL;
 Virginia Beach–Norfolk–Newport N., VA–NC.

Conditions of use. This article may be downloaded from the E&P website for personal research by members of subscribing organisations. This PDF may not be placed on any website (or other online distribution system) without permission of the publisher.